Dwight Look College of ENGINEERING TEXAS A&M UNIVERSITY

ECEN 404
Status Update 1

Group #17: Patent Miner
List of team members:
Christian Casteel
CJ Vines
Kosi Ezewudo
Sponsor: Dr. Reddy Narasimha

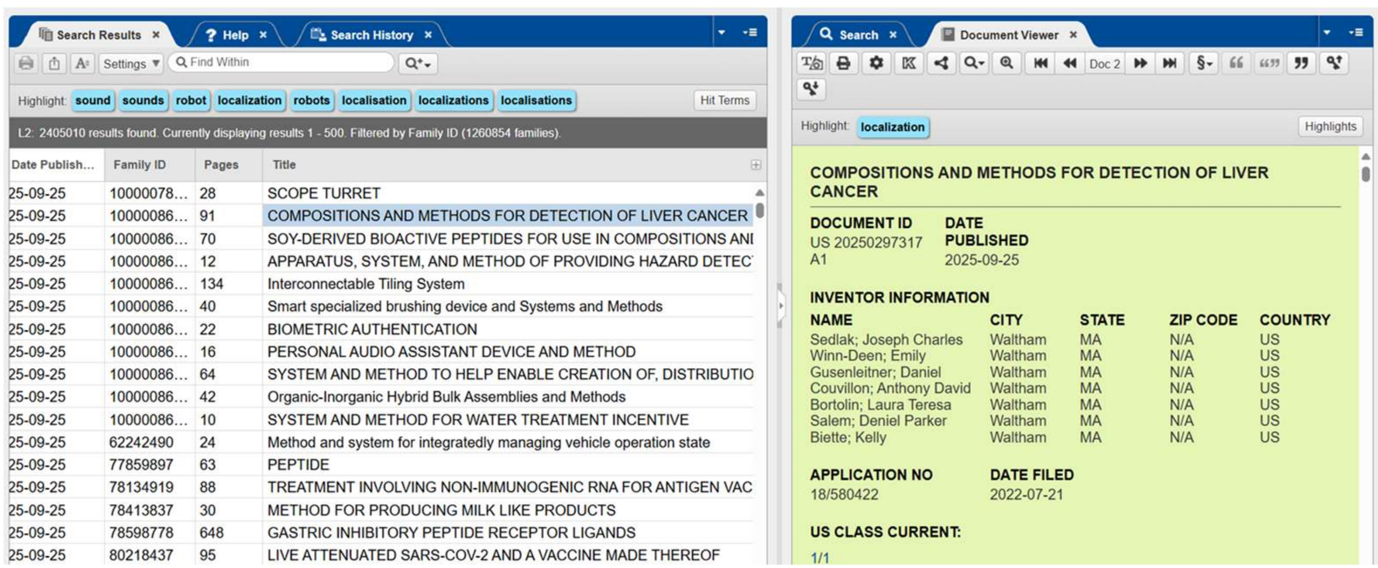
Project Summary

Problem statement:

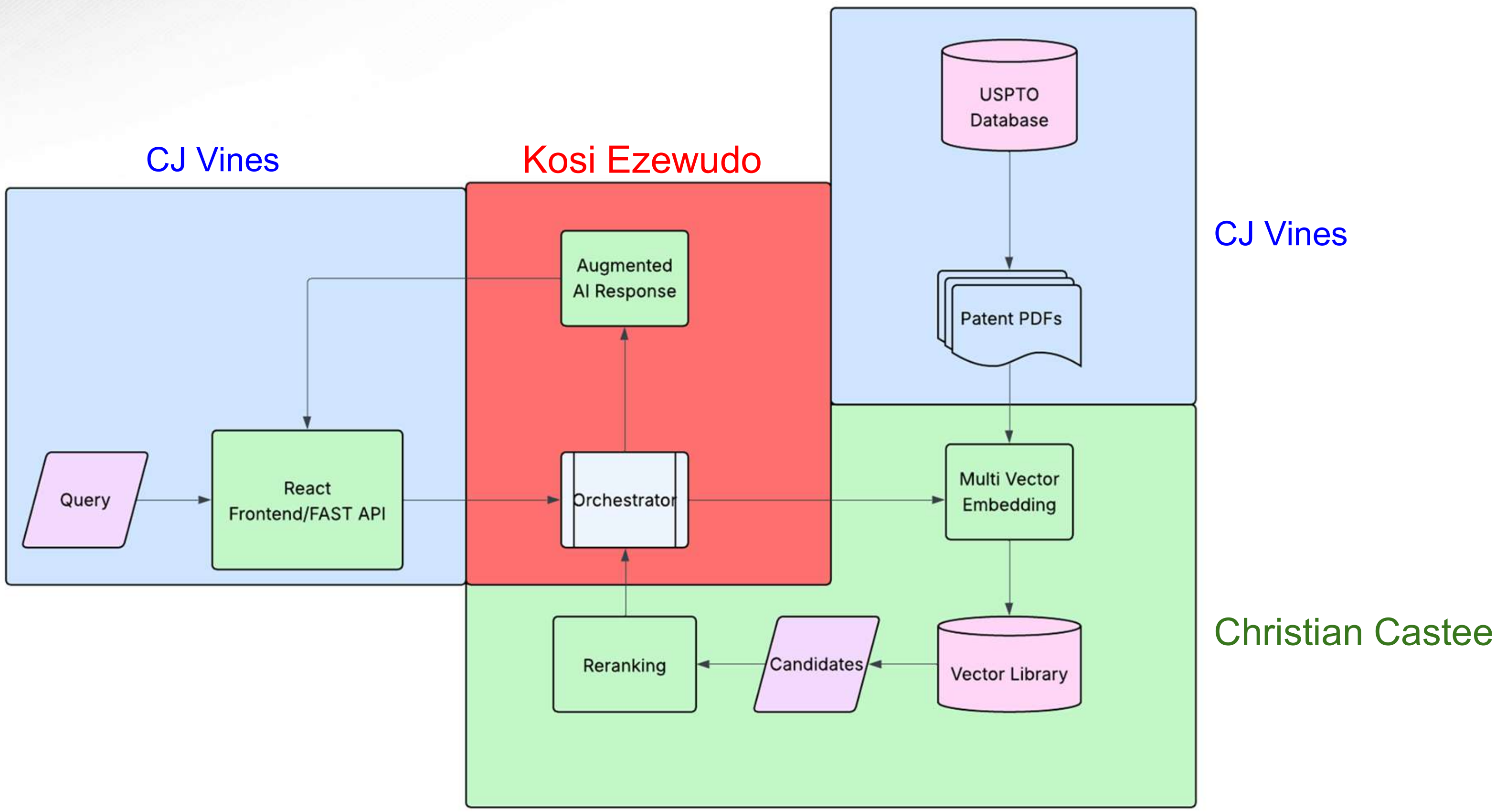
- The current keyword search implemented by the United States Patent and Trademark Office (USPTO) fails to capture contextual meaning, leading to inaccuracy of search results and expensive billing rates for litigation professionals.

Solution:

- Development of a patent database mining tool that optimizes the patent search process by retrieving contextually relevant patents faster and more accurately compared to a traditional database.



System Overview



Project Timeline

1/12

Project Scope

Revisit Subsystems, Define changes, Set dates

1/26

Subsystem Updates Completed

Re-validate using 403 Validation Plan

2/9

Initial Integration + Debugging

Identify potential issues, update ICD on any changes

2/23

Final System First Draft

End-to-end integration completed, begin retrieval tweaking

3/9

Final System Validation

Ensure all functional requirements are met

3/30

Completed System/Additional Features

Major Project Changes

- Project scope: Patent Litigation Tasks
- Usage of ColBERT for patent re-ranking in addition to embedding

- Addition of MUVERA to enhance retrieval speed and vector storage

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Embedding Pipeline and Storage

Christian Casteel

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
• Created pipeline that embeds text from patents in XML format using ColBERT, and stores vectors locally using weaviate • Connected pipeline to USPTO download script	• Verify multi vector storage • Begin implementing ColBERT reranking

Embedding Pipeline and Storage

Christian Casteel


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ements, the table 300 is used by the user device 110 to generate a digital fingerp
rint. In such arrangements, the user device 110 transforms the information contain
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```


Image of a patent chunk’s metadata and a single embedding

Frontend/Preprocessing

CJ Vines

Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
• Created a robust download pipeline with 100% retention of text and figures • Extracted all relevant metadata from patent, which is grouped with each chunk for filtered queries and LLM context • Created a React front end that successfully requests and displays relevant patents with hyperlinks to the full patent download	• Further refine SLURM job script for efficient usage of graphical and processing power • Assisting other subsystems in their implementation

Preprocessing

```
uspto_bulk_data > sampled > sample_05_12405102_Super-sensitive-capacitive-strain-sensor.md > # Super-sensitive ca
# Super-sensitive capacitive strain sensor with electrode fragmentation (Doc 12405102)

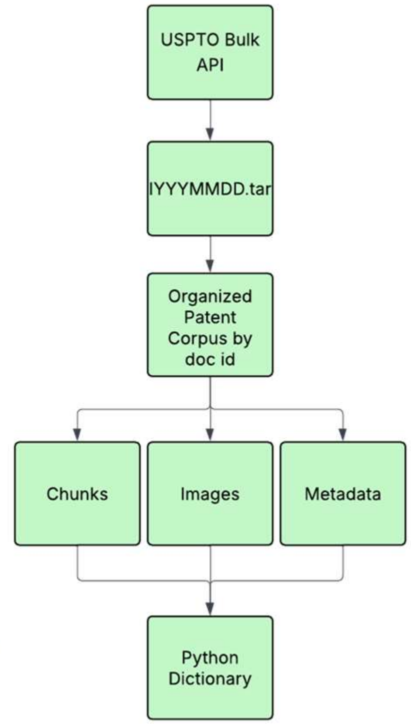
- Doc ID: 12405102
- Title: Super-sensitive capacitive strain sensor with electrode fragmentation
- Filing Date: 20210323
- Classification: G01B 7/22
- Authors: Gilles Lubineau; Hussein Nesser

## Abstract

A capacitive strain sensor configured to measure a strain includes a dielectric layer;
face of the dielectric layer; and a second electrode placed on a second face, opposite
dielectric layer. The first electrode is formed of a single-walled carbon nanotube pape
nanotube paper has plural pre-cracks made according to a pattern.

## Description

This application is a U.S. National Stage Application of International Application No.
23, 2021, which claims priority to U.S. Provisional Patent Application No. 63/002,850,
"HIGHLY SENSITIVE CAPACITIVE STRAIN SENSORS BASED ON FRAGMENTED ELECTRODES," the disclo
herein by reference in their entirety.
```



Sampled 50 random patents from each, reconstructed them in .md format, compared to pdf for accuracy

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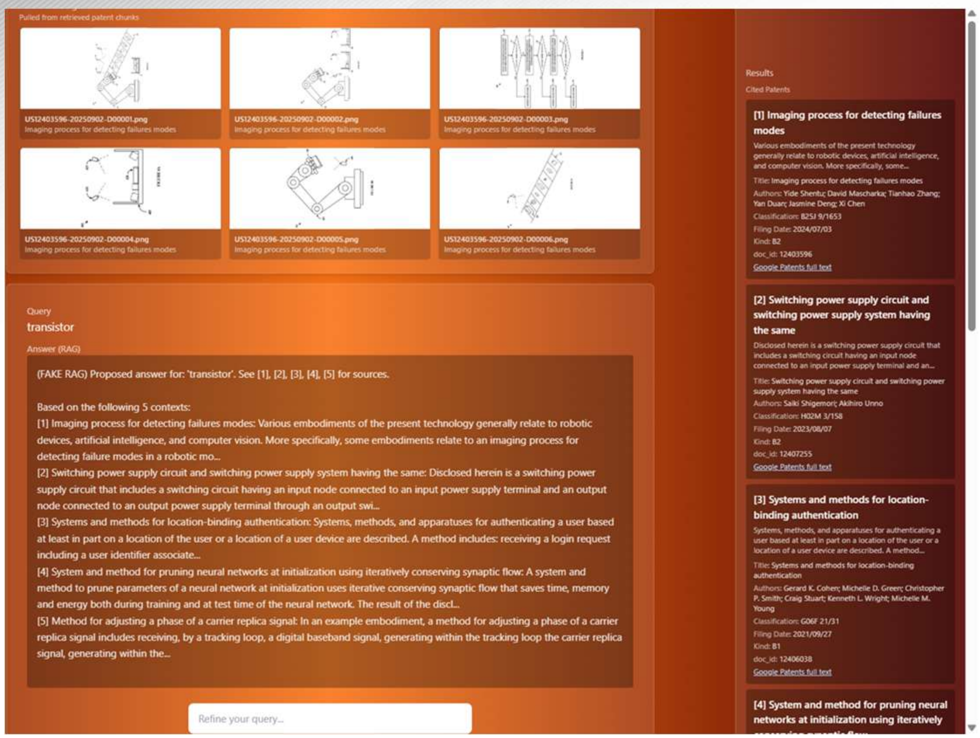
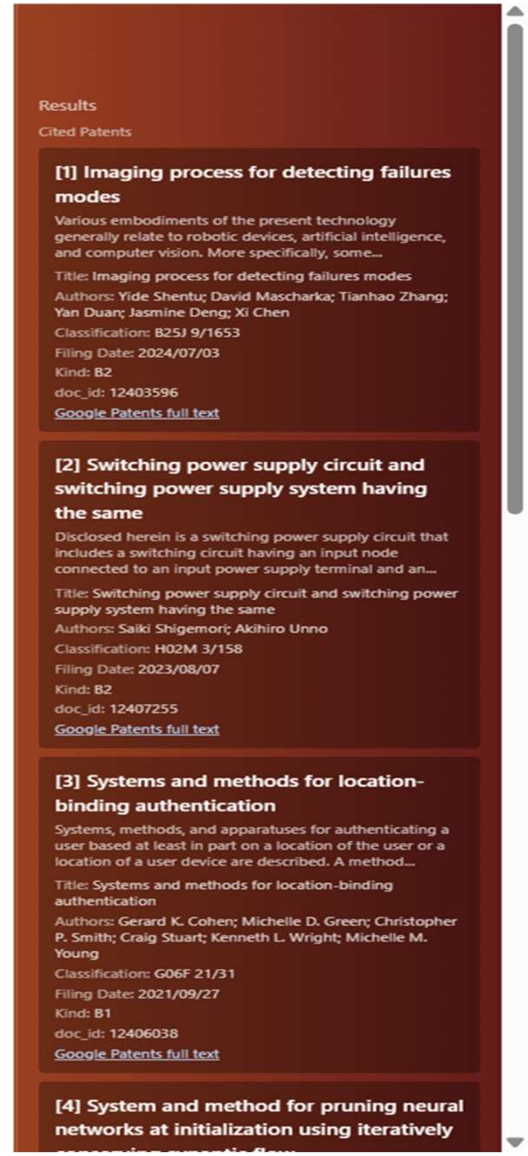
Frontend



- Created pseudo-corpus from golden set of patents to display search functionality and download validation
- Referenced figures in chunks converted to png, displayed to user as well as metadata, links to google patents full text



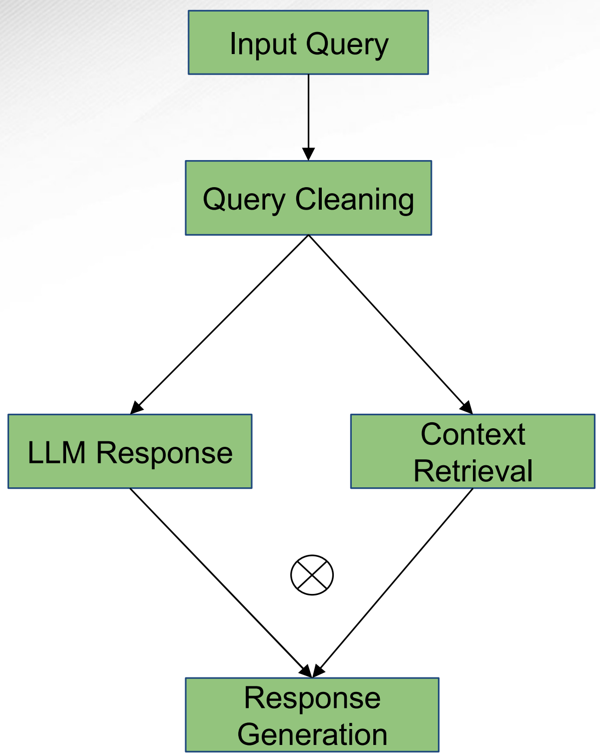
Retrieval Augmented Generation (RAG) Subsystem

Kosi Ezewudo

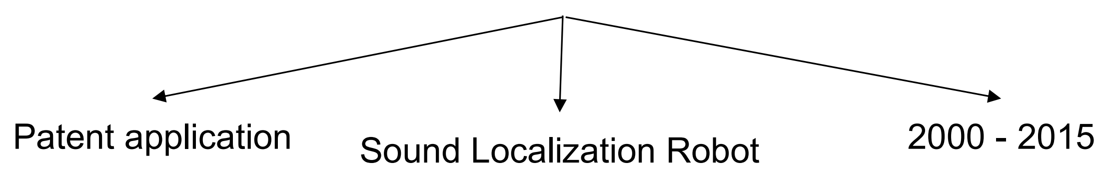
Accomplishments since 403	Ongoing progress/problems and plans until the next presentation
• Implemented a base RAG pipeline that; ○ conditionally executes context retrieval • Implemented a RAG-layer that performs Query Cleaning • Created a stand-alone layer that performs Metadata Extraction	• Finalize and Integrate Metadata extraction layer to RAG pipeline • Finalize subsystem and demonstrate progress

Retrieval Augmented Generation Subsystem

Kosi Ezewudo



Sample Query: Are there any patent applications that were made specifically for sound localization robots between 2000 - 2015?



Execution Plan

	1/12/2026	1/19/2026	1/26/2026	2/2/2026	2/9/2026	2/16/2026	2/23/2026	3/2/2026	3/16/2026	3/23/2026	3/30/2026	4/6/2026	4/13/2026	4/20/2026	Date
Verify Multi Vector Storage															1/19/2026
Finalize Metadata Extraction															1/26/2026
Implement Orchestrator															1/26/2026
Imeplement MUVERA															2/2/2006
Implement ColBERT reranking															2/2/2026
Test Semantic Chunking															2/2/2026
Validate Orchestrator															2/9/2026
Validate Improved Retrieval															2/9/2026
Backend - Orchestrator Integration															2/16/2026
Frontend - Backend Integration															2/23/2026
Full Integration															3/2/2026
Full Validation															4/6/2026
Trial Demo															4/20/2026
Final Demo															4/27/2026

	Not Started
	Behind Schedule
	In Progress
	Complete

Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.2.	API/Pydantic Test	API gives correct response depending on the route taken 100% of the time	Send input query and data of different variations including edge cases like empty queries, queries with special characters, and get requests of the wrong shape	Tested	CJ Vines
3.2.1.1.	Determinism Test	Preprocessing Subsystem resulting chunks consistent across runs on the same dataset	Calculate SHA-256 hash of entire list of dictionaries after chunking and compare hash across runs to ensure download pipeline is deterministic. If any run is inconsistent, the test fails	Tested	CJ Vines
3.2.1.1.	Metadata Extraction	Stored metadata is verified to be at least 99% accurate	Compare metadata after chunking to metadata called from USPTO API, aiming for 99% accuracy	Tested	CJ Vines
3.2.1.4.	Full System Stress Test	Multiple simultaneous API calls are made successfully no data loss within 1ms	Make multiple API calls simultaneously, handle each concurrently in order.	Untested	CJ Vines & Kosi Ezewudo & Christian Casteel
3.2.1.5.	Embedding Reranking Test	"Golden" query - chunk pairs must have ≥ .05 Recall@k, Mean Reciprocal Rank (MRR) ≥ .5, and Normalized Discounted Cumulative Gain (NDCG) of .5	k = 10 target chunks at minimum with an equal number of matching queries. Calculate Recall@k, MRR, and NDCG for dataset. "Golden" set will include 50 hand selected patents.	Untested	Christian Casteel
3.2.1.5.	Embedding Accuracy Test	For a set of chunks, Hit@k of 1 and precision at k above .5. Relevance determined by patent id	Query containing information from multiple patents with similar CPC codes will test retrieval. Calculate Hit@k and Precision@k on top k chunks where k = 10.	Untested	Christian Casteel
3.2.1.1.	Failure Compliance Test	Pipeline can take in incorrect patent data and continue embedding without failure. Also will flag patents with false data	5 fake patents containing empty sections and incorrect CPC codes will be sent through the pipeline.	Untested	Christian Casteel
3.2.1.6.	Metadata Integrity Test	All stored metadata must be 100% consistent with metadata listed on a patent	Select 50 patents, specifically patents with generally unique CPCs and numbers of authors, then compare all metadata fields to the same patents on the USPTO database	Untested	Christian Casteel
3.2.1.1.	Download and Embedding Pipeline Stress Test	A USPTO bulk dataset is downloaded, embedded, and stored without failure	A USPTO bulk dataset download is conducted via fronetend download script. The bulk dataset should contain minimum 1000 patents.	Untested	Christian Casteel & CJ Vines
3.2.1.6.	Query Metadata Extraction	Extract all relevant metadata from user query, and use the metadata to filter patent search	Execute patent search using queries containing various metadata including prior date, patent ID, CPC etc.	Untested	Kosi Ezewudo
3.2.1.3.	Query Cleaning	Removes typos, sentence fragments etc. from user query	Execute patent search using grammatically incorrect queries	Untested	Kosi Ezewudo
3.2.1.4.	Generation Accuracy and Quality Test	Generated responses must have ≥ .8 Answer relevancy scores, ≥ .8 Diversity and Novelty	Query Patent Miner using golden set of patents. Generate responses and calculate Answer relevancy, Diversity and Novelty scores	Untested	Kosi Ezewudo & Christian Casteel & CJ Vines

Thank you!